

The CDF Method in 2D.

Goal: To find the density f_w of a r.v.

$$W = g(Y_1, Y_2)$$

where (Y_1, Y_2) are jointly continuous w/ pdf f_{Y_1, Y_2}

$$\rightarrow: F_w(w) = \mathbb{P}[W \leq w] = \mathbb{P}[g(Y_1, Y_2) \leq w]$$

$$A = \{(y_1, y_2) \in \mathbb{R}^2 : g(y_1, y_2) \leq w\}$$

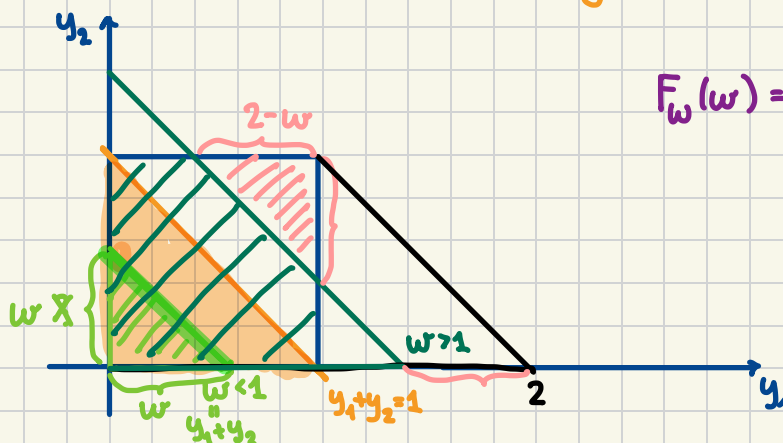
$$= \mathbb{P}[(Y_1, Y_2) \in A]$$

$$F_w(w) = \iint_A f_{Y_1, Y_2}(y_1, y_2) dy_2 dy_1$$

Example. Say (Y_1, Y_2) represent points chosen @ random in the unit square $[0, 1] \times [0, 1] = [0, 1]^2$

$$f_{Y_1, Y_2}(y_1, y_2) = \mathbb{1}_{[0, 1]^2}(y_1, y_2)$$

Define $W = Y_1 + Y_2$, i.e., $g(y_1, y_2) = y_1 + y_2$.

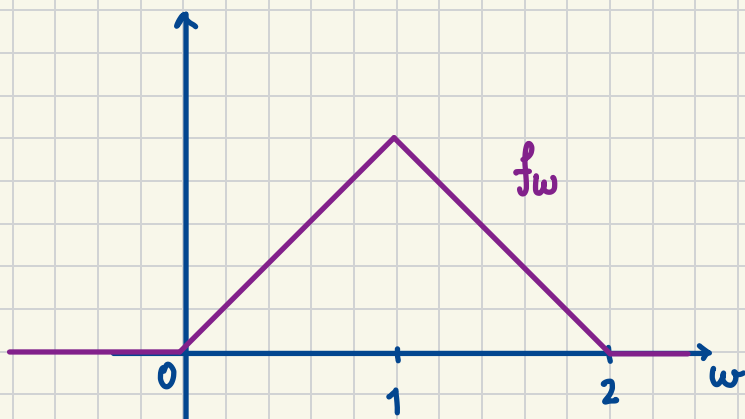


$$F_w(w) = \mathbb{P}[Y_1 + Y_2 \leq w]$$

$$\begin{cases}
 \text{for } w < 0 : F_w(w) = 0 \\
 \text{for } w \in [0, 1) : F_w(w) = \frac{w^2}{2} \\
 \text{for } w = 1 : F_w(w) = \frac{1}{2} \\
 \text{for } w \in (1, 2] : F_w(w) = \boxed{} - \triangle = 1 - \frac{(2-w)^2}{2} \\
 \text{for } w > 2 : F_w(w) = \underline{1}
 \end{cases}$$

$$f_w(w) = \begin{cases} 0 & \text{for } w < 0 \\ \frac{1}{2} \cdot 2w = w & \text{for } w \in [0, 1) \\ 2-w & \text{for } w \in [1, 2] \\ 0 & \text{for } w > 2 \end{cases}$$

!



χ^2 Distribution.

Let $Y \sim N(0, 1)$

Set $W = Y^2$, i.e., $W = g(Y)$ w/ $g(y) = y^2$.

For all $w \leq 0$: $F_W(w) = 0$

For all $w > 0$:

$$F_W(w) = \mathbb{P}[W \leq w] = \mathbb{P}[Y^2 \leq w]$$

$$= \mathbb{P}[-\sqrt{w} \leq Y \leq \sqrt{w}]$$

$$= \mathbb{P}[Y \leq \sqrt{w}] - \mathbb{P}[Y \leq -\sqrt{w}]$$

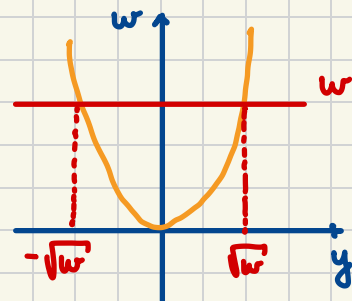


$$= F_Y(\sqrt{w}) - F_Y(-\sqrt{w})$$

$$= \Phi(\sqrt{w}) - \Phi(-\sqrt{w})$$

$$= \Phi(\sqrt{w}) - (1 - \Phi(\sqrt{w}))$$

$$= 2\Phi(\sqrt{w}) - 1$$



By symmetry of $N(0, 1)$:

$$\Phi(-z) = 1 - \Phi(z)$$

$$\text{for } w > 0: f_W(w) = \frac{d}{dw} F_W(w)$$

$$= \frac{d}{dw} (2\Phi(\sqrt{w}) - 1)$$

$$= \cancel{2} \cdot \varphi(\sqrt{w}) \cdot \frac{1}{\cancel{2}\sqrt{w}}$$

$$\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \quad z \in \mathbb{R}$$

$$f_W(w) = \frac{1}{\sqrt{2\pi}} e^{-\frac{w}{2}} \cdot \frac{1}{\sqrt{w}}$$

$$f_W(w) = \frac{1}{\sqrt{2\pi w}} e^{-\frac{w}{2}} \cdot \mathbb{1}_{(0,\infty)}(w)$$

W is said to have the χ^2 -distribution w/
1 degree of freedom.

$$W \sim \chi^2(df=1)$$

More generally, for $Y_1, Y_2, \dots, Y_K$ independent,
the r.v. standard normal r.v.

$$W = Y_1^2 + Y_2^2 + \dots + Y_K^2$$

is said to have the χ^2 -distribution
w/ K degrees of freedom

$$W \sim \chi^2(df=K)$$